Temporal degradation of a news recommender

Working summary, datasets and algorithm — IRE term project, 2026-10-27

Status: planning only. No code has been written and no experiment has been run, so this document contains no results. Hypotheses and expectations are targets, not findings. Deadline 2026-11-20.

1. The project

A news recommender is built from past clicks and articles, but news is replaced within hours. A model built on day *r* meets articles on day *d* that did not exist when it was built. The study measures **how much worse a frozen model gets as days pass, which signal goes stale first, and how often a rebuild pays off.**

Three signals are compared: **word matching** (BM25), **meaning matching** (frozen multilingual MiniLM embeddings) and **click history** (item-item co-click cosine). Reference scorers: recency, popularity, and feed position (NPR only).

	Question
RQ0 (model-free)	How fast do articles age? Share of clicks on articles published after the fit date.
RQ1	How fast does quality fall with days since the fit?
RQ2	Which signal falls first, and why?
RQ3	What rebuild interval balances quality against the cost of rebuilding?

2. Datasets

Figures marked *measured* were read from the local files on 2026-10-26 or 2026-10-27, including the full Adressa archive.

2.1 EB-NeRD large — primary, on disk (measured)

Fact	**Value**
Language, source	Danish, one publisher
Labelled span	15 days, 2023-05-18 to 2023-06-01 (train 18–25 May, validation 25 May–1 Jun)
Volume	1.4–2.0 M impressions and 380–450 k users per day
Impression lists	Yes. Median slate 9 items; 99.4% have exactly one click
Articles	125,541; Danish title and body; `published_time` present (one 1993 outlier filtered)
Click history	Timestamped, back to 27 Apr; median 80–92 clicks per user
Test set	13.5 M impressions, no click labels, unusable for ranking metrics
Main limit	Only 15 labelled days

2.2 NPR, News Portal Recommendations (Globo) — second dataset, downloaded (measured)

Fact	Value
Language, source	Portuguese, G1 portal; CC BY-NC-SA 4.0
Span	118 local days, 2023-01-03 to 2023-05-01 (116 have data)
Impressions	1,402,576, each with exactly one click among about 10 shown items
Users	1,162,802; 71% of impressions non-logged; 82% of users appear once
Articles	148,099; title, topics, publish date; **no body text**
History	Timestamped; median 6 items
Article age	Median clicked article is 0.58 days (13.8 h) old
Risks	87% of impressions from the non-personalised `top-state` list; 41% of clicks on position 1; modest per-day volume; about 9 outage days to drop

Mitigation: report decay per `algo`, pool days into windows, include a position-only reference scorer. Text signals are limited to titles and topics.

2.3 Adressa — optional replication (measured on the full archive)

Fact	Value
Language, source	Norwegian; CC BY-NC-SA 4.0
Span	10 weeks, 1 Jan to 31 Mar 2017 (Norwegian local days; timestamps are UTC, the first is 23:00 UTC on 31 Dec); 90 contiguous days
Scale	**113.6 M events, 4,805,071 users, 81,019 distinct article ids** (the documentation says 27.2 M events, 3.08 M users, 48,486 articles; the difference is unexplained)
Article fields	Title, keywords, author, `category1`, `publishtime` (ISO 8601, UTC), weighted `profile` with entities
Gaps	No article body (on request only); **no impression lists**, so the candidate pool is built from publish time
Per day	1.25–1.61 M events, 237–310 k users, 340–502 k article clicks (3 days sampled)
Status	Converted to parquet (all fields, 3.11 GB, verified) in `data/adressa/parquet/`. Checked: publish time, text fields, 10 contiguous weeks; 34 % of 2 Jan users return on 13 Feb

Rejected: Netflix Prize, AOL, irgroup catalogue, DIGIX, TREC Washington Post, PGNSC, Globo 2017, Kaggle MIND mirrors. Each fails on text, user history, dates or access.

3. Algorithm — explained from the ground up

In one sentence: we measure how much a recommender gets worse the longer it goes without being rebuilt, and which of its parts gets worse first.

1. The plain picture. A recommender learns from last week's clicks. Tomorrow's news is different, so it gets a little worse each day. We want to know how much worse, and when rebuilding it is worth the effort.

2. How we measure it. We take a model built on day *r* and let it rank the articles shown on later days. We also score a model rebuilt every day on the same impressions. The difference between the two is the **decay**.

- *Why the daily model:* it cancels weekday effects. A weekend dip hits both models equally, so it does not read as decay.
- *Why the same impressions:* every comparison is paired, which gives tighter confidence intervals.

3. The three signals, and why they age differently.

Signal	What it knows	Why it may go stale
Word matching (BM25)	Which words readers' past articles used	New stories use new words
Meaning matching (frozen MiniLM)	What articles are about	Least affected; only the topic mix shifts
Click history (co-click)	Which articles were clicked together	A brand-new article has no co-clicks, so it cannot be scored

4. Two versions of "frozen". In **regime A** the whole system, including each reader's profile, is frozen on day *r*. In **regime B** only the learned parts are frozen and profiles keep updating. Without both, the click-history answer would be decided by how we define "frozen".

5. One table answers all three questions. The **decay matrix** holds the decay for every fit day and every later day. From it we read how quickly quality falls (RQ1), which signal falls first (RQ2) and how often to rebuild (RQ3).

6. Safeguards. Only past data is used for each fit. Tests must fail when the future is deliberately leaked. The recency scorer has nothing to freeze, so its decay must come out exactly 0, which checks the harness. Candidates a scorer cannot score tie at the bottom in a seeded shuffle, not in slate order, because slate order may encode the site's own ranking.

7. The precise version.

Notation. r = fit date, d = evaluation day with $d > r$, $M(r, d)$ = metric of the model fitted at r and scored on day d.

1. **Unify** each dataset into one schema: articles, clicks, slates.
2. **Sample once:** a seeded, fixed set of 20,000 impressions per day. Sample ids never change, so every model is scored on identical impressions.
3. **Fit** each scorer for each fit date r on the 7-day sliding window ending at the end of day r: BM25 index, MiniLM article embeddings, co-click item-item matrix. Recency and popularity are fitted the same way.
4. **Score** every sampled slate on each later day d. Candidates a scorer cannot score (for example a new article with no co-clicks) tie at the bottom in a seeded shuffle, never in slate order, because slate order may encode the site's own ranking.

5. **Fill the decay matrix:** M(r, d) for every pair, with AUC, MRR and nDCG@5, plus the paired bootstrap CI and n.
6. **Decay:** Δ(r, d) = M(d−1, d) − M(r, d), the gap to a model rebuilt the previous day. This daily-rebuilt control cancels weekday and difficulty effects.
7. **Two regimes.** A: everything frozen at *r*, including reader profiles. B: only learned parts frozen, profiles live (built from clicks strictly before each impression). Reporting both stops RQ2 being decided by construction.
8. **Rebuild schedule (RQ3):** for interval k in {1, 2, 3, 5, 7, 14} days, $\bar{M}(k)$ is the average of M(r_k(d), d), where r_k(d) is the latest rebuild at or before *d*. Cost is the number of rebuilds and fit seconds.
9. **Diagnostics (RQ2):** coverage of new articles for click history, vocabulary drift for BM25, embedding-space drift for meaning matching.

Leakage boundary. Fitted state uses events up to the end of day *r*. Hybrid weights use an inner split. Mutation tests must fail when the boundary is moved.

Metrics. AUC and MRR primary (EB-NeRD slates are small with one click); nDCG@5 secondary. Always with n and a paired bootstrap CI.

Harness self-check (H4). The recency scorer has nothing to freeze, so its decay gap must come out exactly 0.

Without impression lists (Adressa). The pool at time *t* is every article with `publishtime` ≤ *t* published in the last 48 hours, defined from publish time only. Rank the full pool, not sampled negatives.